Voter2-2K Quick Start Guide

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Here are the steps to perform to setup your Voter2-2K for the first time from factory defaults. You will need the Voter2 Users Guide for reference. For more info, see the Basic Setup on page 11 of the Users Guide.

1. Install the Voter2-2K board and power up the station.
    - The board can be installed in the MTR2000 Option Slot 1 or 2, either the middle or bottom slot.
    - The MTR2000 Wireline card must be installed in the top slot for proper Full-Duplex operation of the MTR2000 station. It is not used by the Voter2.
    - Plug in the line cord of the MTR2000 and wait for it to power up. It may indicate a red failure LED at this point. We'll fix that in the next step.
2. Plug in the RSS cable, run RSS and edit the board configuration.
    - Plug in the RSS cable to the MTR2000 RSS connector. Connect the other end to your computer running Motorola MTR2000 RSS.
    - Run the RSS software.
    - In the RSS software, click **File | Read Station**.
    - On the **Station Configuration** dialog that is displayed, click **Edit Config**.
    - Click on **Wireline Board Type** and select the appropriate **4-Wire** board type.
    - Click on **Option Board Type** and select **Aux I/O**.
    - Click Ok. Click **File | Write Station**.
3. Plug in the Voter2 USB-C Console cable and connect your terminal program.
    - Plug a USB-C cable into the console connector of the Voter2-2K. Connect the other end to your computer.
    - Run a terminal emulator program, e.g. Tera Term or Putty.
    - Set the terminal emulator's serial port baudrate to 115200, 8, N, 1.
4. On the terminal emulator, login to the Voter2 with the root login.
    - Press enter until the prompt says `Username`
    - Enter a `Username` of `root`
    - Enter a `Password` of `guru`
5. Setup the radio interface.
    - `help status` – see the list of status display options.
        - There are a number of summary status screens available to you.
    - `help show` – view the list of configuration options and abbreviations.
    - The `show` command is used to view individual configuration options. Type `show xxxx` where `xxxx` is the name of the configuration option or its abbreviation.
    - The `set` command is used to set individual configuration options. Type `set xxxx yyyy` where `xxxx` is the name of the configuration option or its abbreviation and `yyyy` is the value to set it to.

- `set gpsdev liv4f` – set up GPS operation for the ST Teseo-LIV4F module that is preinstalled on your board.
- `set rssimode analog` – set up the RSSI source as the MTR2000 station-generated RSSI output for voting.
- `set rxmode cor` – we're going to use the MTR2000 COR logic output for the balance of setup. There's a note below about resetting it to `ctcss` when setup is complete.
- Note: unlike the original Voter board, the Voter2 does not have a built-in squelch and instead depends on the MTR2000 squelch for its COR.
- Notice that there is an asterisk displayed at your prompt, like this: `root*>`
  This reminds you that you have made changes that are not yet permanently saved in nonvolatile memory. Use the command `settings save` to save your changes to nonvolatile memory.

6. Plug in your GPS 3.3V Active Antenna.
   - The GPS receiver is searching for satellites. A flashing yellow GPS LED indicates decoding is in progress. A solid yellow GPS LED indicates sufficient satellites have been decoded.
   - `status gps` -- shows you the gps connection information and satellites received.
7. Setup your `local mode` operation.
   - `set local mode repeater` – you will use local mode to set the transmit audio level.
8. Perform your Rx and Tx Audio Level Calibration.
   Follow the instructions in Appendix D of the Users Guide, page 36.
   - Make sure that you are NOT generating CTCSS with your service monitor.
9. Perform your RSSI Calibration.
   Follow the instructions in Appendix E of the Users Guide, page 39.
   - Notice that there is an asterisk displayed at your prompt, like this: `root*>`
     This reminds you that you have made changes that are not yet permanently saved in nonvolatile memory. Use the command `settings save` to save your changes to nonvolatile memory.
10. Perform your Squelch Calibration.
    Follow the instructions in Appendix F of the Users Guide, page 43.
11. Plug in your network connection.
    - Plug your network cable into the shielded connector. This is the network connector.
12. Setup your network connection.
    If you are using DHCP, the Factory Defaults will work fine.
    - `set myipmode dhcp` – set this configuration option to `dhcp`.
    - `status ethernet` will show you the connection information.

    If you are setting up your own IP address, set the following options:
    - `set myipmode static` – set this option to `static`.
    - `set mystaticip <ip address>` – set your IP Address, e.g. 192.168.20.220
    - `set mystaticsn <subnet>` – set your Subnet Mask, e.g. 255.255.255.0
    - `set mystaticgw <gateway ip address>` – set your Gateway Address, e.g. 192.168.20.1
    - `set mystaticdns <dns ip address>` – set your DNS Address, e.g. 192.168.20.1
    - `status ethernet` will show you the connection information.

13. Setup your AllstarLink (DIAL) connection.

    Configure these options.

    - `set dial ipmode static` – set this configuration option to `static` or `dns`.
    - `set dial port 1667` – set to `667` for pre-ASL3; set to `1667` for ASL3.
    - `set dial staticip <ASL ip address>` -- set the ip address of the voter server.
    - `set dial fqdn <fully qualified domain name>` – used when `ipmode` is set to `dns`.
    - `set dial voterpwd <password>` – set the voter-unique password defined as the secret in the voter client definition of voter.conf.
    - `set dial hostpwd <password>` – set the host password defined in the general stanza of voter.conf.
    - `status dial` -- shows the connection status.

14. Set `rxmode` as required.
    - In step 5, you set `rxmode` to `cor`. This setting simplified the setting of audio levels.
    - `set rxmode xxxx` as required from this list: `cor`, `ctcss`, `corctcss`